QUIZ 01 PRACTICE PROBLEMS – Solutions

STAT 1110: Section S18 – Fall 2026 with Dr. Ethan P. Marzban

⚠ Caution

These problems are not meant to be exactly indicative of what will show up on the Quiz on Monday. Rather, they are designed to give you a sense of the style of multiple choice questions I typically ask. As a reminder, the quiz on Monday will be 15 questions.

Use the following setup for Questions 1 through 10:

Quinn, an avid enthusiast of cars, has visited several dealerships around San Luis Obispo. For each of the 90 cars she observed, she made note of the following attributes:

- **Weight**: the car's weight (in tons)
- **MPG**: the car's advertised mileage per gallon (i.e. how many miles the car is reported to be able to travel on one gallon of fuel)
- **Mileage**: the mileage recorded on the cars odometer (i.e. how many miles the car has driven up until now)
- **Make**: the make of the car (e.g. "Ford," "Honda," etc.)
- **Model**: the model of the car (e.g. "sedan," "SUV," etc.)
- **Electric**: whether the car was Electric, Hybrid, or Gas (encoded as "1" for Electric, "2" for Hybrid, or "3" for Gas)

1. Describe an observational unit in the context of this dataset.

(**A**) Quinn

(**B**) One car dealership

(C) One car

(**D**) The weight of one car

(**E**) San Luis Obispo

Solution: We are told that Quinn collected various pieces of information (weight, MPG, etc.) from the individual *cars* she encountered - as such, an observational unit in this dataset is **one car**.

2. Is the **Electric** variable categorical or quantitative?

(**A**) Categorical **(B) Quantitative**

Solution: Even though values have been encoded using numbers (1, 2, and 3), it makes no interpretive sense to perform arithmetic operations on values ("1" + "2" really means "Electric" + "Hybrid," which makes no sense!)

3. What type of plot would best display the distribution of the **Make** variable? Select only one answer choice below.

   (A) Bargraph/barplot

   (**B**) Histogram

   (**C**) Scatterplot

   (**D**) Dotplot

   (**E**) More than one of the answer choices above is correct

   Solution: **Make** is a categorical variable, and to display the distribution of a single categorical variable we use a **bargraph/barplot**

4. What type of plot would best display the relationship between the **Weight** and **MPG** variables? Select only one answer choice below.

   (**A**) Bargraph/barplot

   (**B**) Histogram

   (C) Scatterplot

   (**D**) Dotplot

   (**E**) More than one of the answer choices above is correct

   Solution: **Weight** and **MPG** are both quantitative variables. To display the relationship between two quantitative variables, we use a **scatterplot**.

In addition to the original setup, use the following for questions 5 - 10:

Shown below in Figure (1) is the distribution of the **MPG** variable:

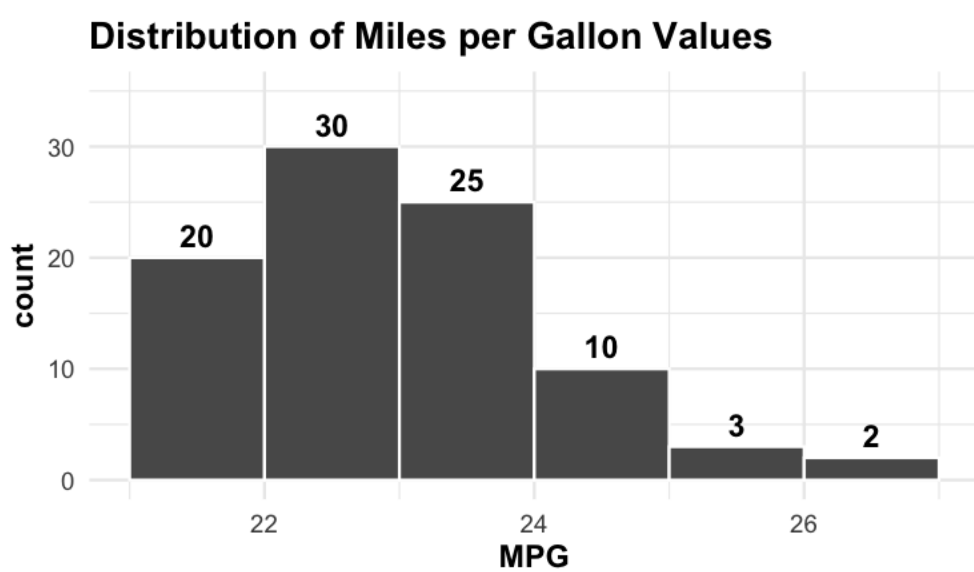

Figure 1: Distribution of MPG values of the 90 cars Quinn observed. Bins range from 21 and 27 in increements of 1; counts are 20, 30, 25, 10, 3, and 2, respectively.

5. Is the distribution of MPG values unimodal, bimodal, or something else?

(A) Unimodal (**B**) Bimodal (**C**) Something Else

Solution: There is only one "peak," meaning the distribution is unimodal.

6. Are there any outliers in the graph?

(**A**) Yes **(B) No**

Solution: There are no points that are too distant from the bulk of the data, meaning there are no outliers.

7. Is the distribution of MPG values symmetric, or skewed?

(**A**) Symmetric **(B) Right-Skewed** (**C**) Left-Skewed

Solution: Because the tail extends to the right, we say the distribution is right-skewed.

8. Suppose we decided that the distribution of MPG values is left-skewed (which is not necessarily the correct answer to Question (7) above! Would the mean and the median necessarily be equal?

(**A**) Yes, the mean and median would be roughly equal.

(**B**) No, the mean would differ from the median.

Solution: The mean and median are only roughly equal when the distribution is roughly symmetric.

9. Approximately what proportion of cars in Quinn's dataset had MPG values between 21 and 24?

(**A**) 22.2222%

(**B**) 27.7778%

(**C**) 33.3333%

(D) 83.3333%

(**E**) 100.0000%

Solution: The number of cars with MPG values between 21 and 24 is 20 + 30 + 25 = 75. The total number of cars Quinn observed is 90 (which is given in the problem statement, and could also have been computed by summing up all the counts in the histogram in Figure 1). Hence, the proportion of cars in Quinn's dataset with MPG values between 21 and 24 is **75/90 = 83.3333%**.

10. The mean MPG value of the 90 cars Quinn observed is around 23. A new (91st) car is added to the dataset; this one has a MPG of 45. What is the mean of all 91 cars (the original 90 Quinn observed, and the new 91st car with MPG of 45)?

(**A**) 19.3928 mpg

(**B**) 23.0000 mpg

(C) 23.2418 mpg

(**D**) 23.5000 mpg

(**E**) 68.0000 mpg

Solution: The sum of the MPG's of the 90 cars Quinn observed can be calculated by taking the mean MPG and multiplying by 90 (the number of cars Quinn observed):

$$(\text{sum}) = 23 \times 90 = 2070$$

The sum of all 91 car's MPG values is then

$$(\text{new sum}) = (\text{old sum}) + 45 = 2070 + 45 = 2115$$

and so the mean MPG among the 91 cars is

$$\frac{2115}{91} \approx 23.2418$$

The following questions are unrelated

11. Can a standard deviation ever be negative (i.e. less than zero)?

(**A**) Yes **(B) No**

Solution: As indicated on the formula sheet (and as you saw in Section P2 of the textbook), the SD represents a "typical distance in absolute value from the mean." This means, by definition, that an SD cannot be negative.

12. Can a proportion ever be negative (i.e. less than zero)?

(**A**) Yes **(B) No**

Solution: A proportion is the ratio of two counts. Counts can never be negative; therefore we can never have a negative proportion.

13. Can a proportion ever be equal to zero?

(A) Yes (**B**) No

Solution: As an example, consider the Palmer Penguins dataset, in which Dr. Gorman only observed Adélie, Chinstrap, and Gentoo penguins: the proportion of, say, Emperor penguins is 0 since there were zero Emperor penguins included in the dataset.

14. Can a median ever be negative?

(A) Yes (**B**) No

Solution: There are many cases where the median may be negative! For example, suppose we collected temperatures in Centigrade at the North Pole over several days; I'd wager that most of the temperature readings (if not all of them) would be negative, and therefore their median would also be negative.

15. Consider the following student's argument: if all the values in a list of numbers are negative, then the IQR will also be negative. Is the student correct? Why or why not?

(**A**) The student is correct; furthermore, the only way for the IQR to be negative is if all observed values are negative.

(**B**) The student is correct, however the IQR may also be negative even if only some (not all) observed values are negative.

(C) The student is incorrect, as Q_3 will always be larger than or equal to Q_1, meaning the IQR will always be nonnegative (i.e. it can only ever be zero or positive).

Solution: Q_1 is defined to be the median of values less than the median and Q_3 is defined to be the median of values greater than the median; therefore, Q_3 will in fact always be no smaller than Q_1; hence the IQR will always be no smaller than zero. Therefore, answer choice C is correct.

16. A businesswoman calculates that the median cost of the five business trips that she took in a month is \$600 and concludes that the total cost must have been \$3000. Does this conclusion necessarily follow?

(**A**) Yes, it follows; the median is defined to be the sum of all observations divided by the total number of observations, meaning the sum of all observations is equal to the median times the number of observations. $\$600 \times 5 = \$3000$, meaning the businesswoman's conclusions are correct.

(**B**) No, because the businesswoman made an arithmetic error: the median is defined to be the sum of all observations divided by the total number of observations, meaning the sum of all observations is equal to the median times the number of observations. But, \$600 $\times$ 5 is not equal to \$3000, meaning the businesswoman's conclusions are incorrect.

(C) No, because the businesswoman has used an incorrect formula for the median: if the <u>mean</u> cost of her five business trips were \$600 then her total expenditure would have been \$3000. There is no direct connection between the median and the sum.